

Chapter 4 Nominal and Effective Interest Rates

- **Illustrative Example**

- You placed \$100 in a saving account for one year at an interest rate of 1% *per month*.
- Calculate the amount of interest and annual interest rate.
 - $F = P(1+i)^n = 100 \times 1.01^{12} = \112.68 . The interest earned is $112.68 - 100 = \$12.68$.
 - The annual interest rate is $12.68/100 = 12.68\%$.
- We say that the *effective* annual interest rate is 12.68%.
- Or, the interest rate is 12% per year, compounded monthly.
- That is, the effective annual interest that corresponds to a 12% *nominal* annual interest, *compounded* monthly is 12.68%.

- **Nominal interest rate**

- A nominal interest rate is an interest rate that does not include any consideration of compounding.
- Means “in name only”, “not the true, effective rate.” E.g.,
- 12% per year, compounded monthly
 - 12% is NOT the true effective rate (per year)
 - 12% represents the nominal rate
- Nominal interest rate is commonly referred to as “APR” (annual percentage rate).

- **Effective interest rate**

- Effective interest rate is the actual rate that applies for a stated period of time.
- It takes into account the effect compounding of interest
- Effective interest is stated in the following form:

r (per year), compounded every CP .

- It involves two parameters
 - The annual nominal rate r .
 - The compounding period, CP , the time where interest applies
 - E.g.,
 - Daily compounding, $CP = 1 \text{ day} = 1/365 \text{ year}$.
 - Weekly compounding, $CP = 1 \text{ week} = 1/52 \text{ year}$.
 - Monthly compounding, $CP = 1 \text{ month} = 1/12 \text{ year}$.
 - Quarterly compounding, $CP = 3 \text{ months} = 1/4 \text{ year}$.
 - Semiannual compounding, $CP = 6 \text{ months} = 1/2 \text{ year}$.
- The effective rate is called APY (annual percentage yield).

- **Factors under m -time-a-year compounding**

- Under compounding over a period $CP = 1/m \text{ year}$ (e.g., $CP = 1/12 \text{ year} = 1 \text{ month}$), and at a nominal interest rate r , a present current P is equivalent after k periods (e.g. months) to

$$F = P(1+r/m)^k \Rightarrow (P/F, r, m, k) = (1+r/m)^k .$$

- Similarly, F dollars after k periods are equivalent to

$$P = F / (1+r/m)^k \Rightarrow (F/P, r, m, k) = 1 / (1+r/m)^k .$$

- **Computing the effective interest rate**

- Note that the effective interest rate per CP is r / m , where $m = 1/CP$, with CP given in fraction of a year, is the number of times interest is compounded per year.
- E.g., with monthly compounding, $m = 12$, and a nominal rate of 12% translates into an effective monthly rate of 1% .
- With semiannual compounding, $m = 2$, and a nominal rate of 12% translates into an effective semiannual rate of 6% .
- Then, \$1 is equivalent to $(1 + r / m)^m$ after 1 year.
- The effective annual rate is such that $1 + i = (1 + r / m)^m$. I.e.,

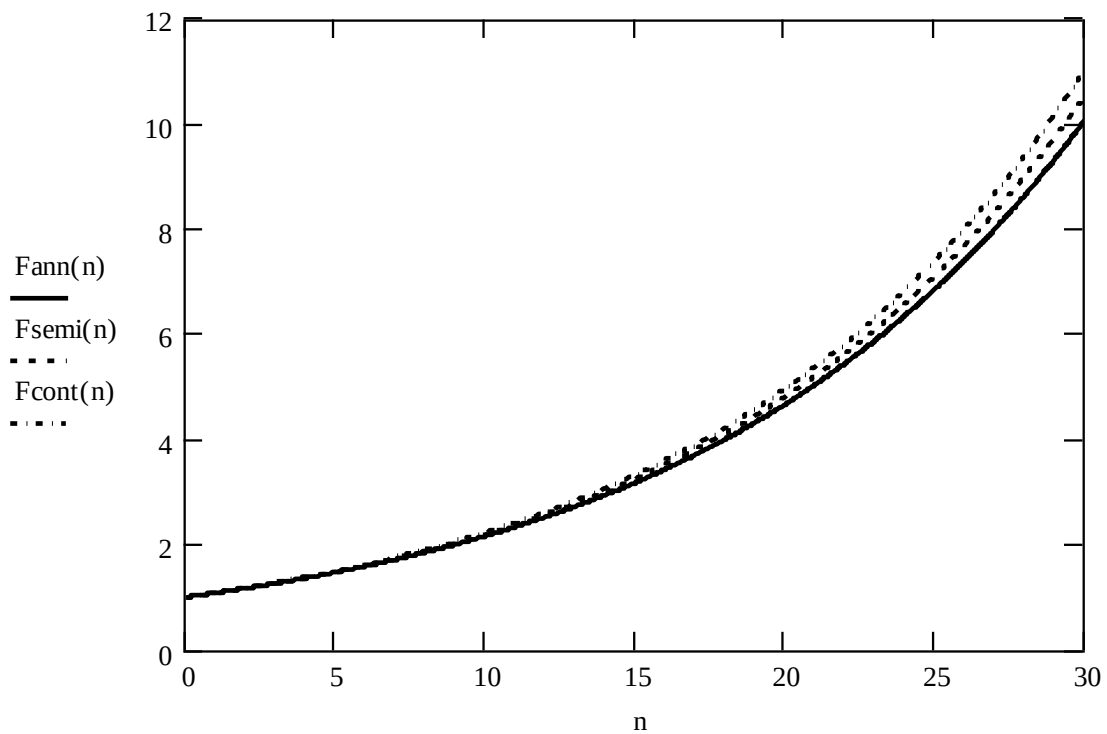
$$i = \left(1 + \frac{r}{m}\right)^m - 1.$$

- **Continuous compounding**

- If the compounding period, CP , is too small, $CP \rightarrow 0$, the number of compounding times gets too large, $m \rightarrow \infty$.
- This situation is known as “continuous compounding.”
- By noting that there are mt compoundings over a time t expressed in years, the F/P factor over time t , under continuous compounding, is $\lim_{m \rightarrow \infty} (1 + r / m)^{mt} = e^{rt}$
- Similarly, under continuous compounding, the effective annual rate is $i = e^r - 1$.
- Continuous compounding is often assumed in quantitative finance as it simplifies the analysis.

- **Example: \$1 invested at a nominal rate of 8%**

Year	$m = 1$	$m = 2$	$m = 4$	$m = 12$	$m = \infty$
1	1.080	1.082	1.082	1.083	1.083
2	1.166	1.170	1.172	1.173	1.174
3	1.260	1.265	1.268	1.270	1.271
4	1.360	1.369	1.373	1.376	1.377
5	1.469	1.480	1.486	1.490	1.492
6	1.587	1.601	1.608	1.614	1.616
7	1.714	1.732	1.741	1.747	1.751
8	1.851	1.873	1.885	1.892	1.896
9	1.999	2.026	2.040	2.050	2.054
10	2.159	2.191	2.208	2.220	2.226



- **Interest rate that varies with time**

- In practice, interest rate may vary from one period to the other.
- In particular, it is often *expected* that the interest rate will *increase* with time.
- If the interest rates in periods, 1, ..., n are $i_1, \dots, i_n$. Then, the future worth after n periods, F , of a present amount P is

$$F = P(1+i_1)(1+i_2)\dots(1+i_n) = P \prod_{t=1}^n (1+i_t).$$

- The rates $i_1, \dots, i_n$ are known as *short rates*.
- The short rate i_t represents the expected 1-year rate after t years.